

# Setup & Prerequisites

This vignette covers everything you need **before** using **snowflakeR** – system requirements, Python setup, authentication, and environment-specific configuration. For usage, see `vignette("getting-started")`.

## Quick reference

| Requirement               | Version                       |
|---------------------------|-------------------------------|
| R                         | $\geq 4.1.0$                  |
| Python                    | $\geq 3.9$ (3.11 recommended) |
| snowflake-ml-python       | $\geq 1.5.0$                  |
| snowflake-snowpark-python | (latest)                      |
| reticulate                | $\geq 1.34$                   |

## 1. System requirements

### R

Install R  $\geq 4.1.0$  from CRAN or via your system package manager. **snowflakeR** uses the base pipe (`|>`) and other features introduced in R 4.1.

### Python

**snowflakeR** uses **reticulate** to call the Snowflake ML Python SDK under the hood. You need Python  $\geq 3.9$  available on your system. We recommend Python 3.11 for best compatibility with **snowflake-ml-python**.

### Conda (recommended) or pip

We strongly recommend using **conda** (or **micromamba**) to manage the Python environment. This avoids version conflicts between **snowflake-ml-python**, **pandas**, **pyarrow**, and other dependencies. **pip** works fine too if you prefer **virtualenvs**.

## 2. Installing snowflakeR

```
# Install from GitHub
# install.packages("pak")
pak::pak("Snowflake-Labs/snowflakeR")
```

## 3. Setting up the Python environment

Choose **one** of the following approaches.

### Option A: Automatic (recommended for most users)

```
library(snowflakeR)
sfr_install_python_deps()
```

This creates a conda environment called `r-snowflakeR` with Python 3.11 and all required packages. After running this once, `reticulate` will automatically find and use the environment.

#### Option B: Conda (manual)

```
conda create -n r-snowflakeR python=3.11 -y
conda activate r-snowflakeR
pip install snowflake-ml-python snowflake-snowpark-python
```

Then tell R to use it:

```
# In your .Renviron or at the top of your script
Sys.setenv(RETICULATE_PYTHON = "~/conda/envs/r-snowflakeR/bin/python")

# Or in R:
reticulate::use_condaenv("r-snowflakeR")
```

#### Option C: pip / virtualenv

```
python3.11 -m venv ~/.virtualenvs/r-snowflakeR
source ~/.virtualenvs/r-snowflakeR/bin/activate
pip install snowflake-ml-python snowflake-snowpark-python

reticulate::use_virtualenv("~/virtualenvs/r-snowflakeR")
```

#### Option D: System Python

If you install the Snowflake packages into your system Python:

```
pip install snowflake-ml-python snowflake-snowpark-python
```

No extra `reticulate` configuration needed – it will find your system Python automatically.

## 4. Verifying the setup

After installing `snowflakeR` and setting up Python, verify everything works:

```
library(snowflakeR)
sfr_check_environment()
#> R version: "4.3.1"
#> v reticulate available
#> Python: "/path/to/python"
#> v snowflake-ml-python available
```

This checks:

- R version
- `reticulate` availability
- Python discovery
- `snowflake-ml-python` import
- `snowflakeauth` availability (optional)

## 5. Authentication

`snowflakeR` supports multiple authentication methods. Choose the one that fits your environment.

## connections.toml (recommended for local use)

Create ~/.snowflake/connections.toml:

```
[default]
account = "xy12345.us-east-1"
user = "MYUSER"
authenticator = "externalbrowser"
warehouse = "COMPUTE_WH"
database = "MY_DB"
schema = "PUBLIC"

[production]
account = "xy12345.us-east-1"
user = "SVC_USER"
private_key_path = "~/.snowflake/rsa_key.p8"
warehouse = "PROD_WH"
database = "PROD_DB"
schema = "ML"
```

Then connect with:

```
conn <- sfr_connect() # uses [default]
conn <- sfr_connect(name = "production") # uses [production]
```

If snowflakeauth is installed, snowflakeR uses it for credential resolution. If not, snowflakeR reads the TOML file directly.

## Key-pair authentication

Generate a key pair:

```
openssl genrsa 2048 | openssl pkcs8 -topk8 -inform PEM -out rsa_key.p8 -nocrypt
openssl rsa -in rsa_key.p8 -pubout -out rsa_key.pub
```

Register the public key with your Snowflake user, then:

```
conn <- sfr_connect(
  account = "xy12345.us-east-1",
  user = "MYUSER",
  private_key_file = "~/.snowflake/rsa_key.p8"
)
```

This is the recommended method for CI/CD and automated workflows.

## External browser (SSO)

```
conn <- sfr_connect(
  account = "xy12345.us-east-1",
  user = "MYUSER",
  authenticator = "externalbrowser"
)
```

Opens your browser for SSO. Good for interactive local sessions.

## Workspace Notebooks (auto-detect)

No configuration needed. See the Workspace Notebooks section below.

## 6. Environment-specific setup

### RStudio / Posit Workbench

This is the most common setup. After steps 2-3 above, configure `reticulate` to find your Python:

**Option 1: .Renviron file** (recommended – persists across sessions)

Create or edit `~/.Renviron`:

```
RETICULATE_PYTHON=~/.conda/envs/r-snowflakeR/bin/python
```

Restart R after editing.

**Option 2: RStudio Python settings**

Go to **Tools > Global Options > Python** and select the `r-snowflakeR` interpreter.

**Option 3: Per-script**

```
reticulate::use_condaenv("r-snowflakeR")
library(snowflakeR)
```

**Note:** `reticulate` locks the Python version on first use in an R session. Make sure to configure Python *before* calling any `snowflakeR` functions.

### VS Code

1. Install the R Extension for VS Code.
2. Set the Python path in your workspace `.Renviron` or in VS Code settings:

```
{
  "r.rterm.option": ["--no-save", "--no-restore"],
  "python.defaultInterpreterPath": "~/.conda/envs/r-snowflakeR/bin/python"
}
```

3. For Jupyter notebooks in VS Code with an R kernel, install `IRkernel`:

```
IRkernel::installspec()
```

The `.Renviron` approach works the same as in RStudio.

### Jupyter Notebooks (local)

There are two ways to use `snowflakeR` in local Jupyter notebooks:

**Option A: R kernel (native R)**

Install `IRkernel` in your R environment:

```
install.packages("IRkernel")
IRkernel::installspec(user = TRUE)
```

Then create a notebook with the R kernel. `snowflakeR` code runs natively:

```
library(snowflakeR)
conn <- sfr_connect()
```

Make sure the R kernel can find Python by setting `RETICULATE_PYTHON` in `~/.Renviron` before launching Jupyter.

**Option B: Python kernel + rpy2**

This mirrors the Workspace Notebook pattern. Install `rpy2` in your Python environment:

```
pip install rpy2
```

In a Python notebook:

```
%load_ext rpy2.ipython
```

```
%%R
library(snowflakeR)
conn <- sfr_connect()
```

## Snowflake Workspace Notebooks

Workspace Notebooks are managed Jupyter environments inside Snowflake. They run a **Python kernel** – R is available via `rpy2` and `%%R` magic cells.

The R environment is installed by the `sfnb-multilang` toolkit, which handles micromamba, R, `rpy2`, and package installation. The `snowflakeR` package ships demo notebooks in `inst/notebooks/` – find them with:

```
system.file("notebooks", package = "snowflakeR")
```

Upload the notebook(s) you need to your Workspace and follow the instructions in the first cell. Key notebooks:

| Notebook                                      | Description                                     |
|-----------------------------------------------|-------------------------------------------------|
| <code>workspace_quickstart.ipynb</code>       | Connection, config, queries, ggplot2            |
| <code>workspace_model_registry.ipynb</code>   | Log, deploy, and serve R models                 |
| <code>workspace_feature_store.ipynb</code>    | Feature Store: entities, views, training data   |
| <code>workspace_forecasting_demo.ipynb</code> | Time series forecasting with forecast + ggplot2 |

## Key differences from local:

| Aspect            | Local                                    | Workspace                                                             |
|-------------------|------------------------------------------|-----------------------------------------------------------------------|
| R kernel          | Native R                                 | Python + <code>%%R</code>                                             |
| Python/R env      | You manage                               | <code>setup_notebook()</code> from <code>sfnb_setup.py</code>         |
| Authentication    | <code>connections.toml</code> / key-pair | Auto-detected (SPCS OAuth)                                            |
| Package install   | <code>install.packages()</code>          | YAML config + tarballs or <code>pak</code>                            |
| Execution context | Usually set by connection                | <code>setup_notebook()</code> reads config or session defaults (docs) |
| Plotting          | Native plot pane                         | <code>%%R -w W -h H + print(p)</code>                                 |
| Session lifecycle | Persistent                               | Re-setup each session (~1-2 min)                                      |

## Setup steps:

1. **Install R and register `%%R` magic** (run once per session, ~1-2 minutes):

```
# Python cell
from sfnb_setup import install_r
install_r()
```

Each notebook directory includes `sfnb_setup.py`, which auto-detects the environment and installs `sfnb-multilang` from PyPI if needed. For advanced configurations (ADBC, custom packages, Scala/Julia), pass keyword arguments or a YAML config: `install_r(config="configs/snowflaker.yaml")`.

2. **Install and load `snowflakeR`:**

```

%%R
options(repos = c(CRAN = "https://cloud.r-project.org"))

if (nzchar(Sys.getenv("SNOWFLAKER_PATH"))) {
  install.packages(Sys.getenv("SNOWFLAKER_PATH"), repos = NULL, type = "source")
} else {
  install.packages("pak", type = "source", quiet = TRUE)
  pak::pak("Snowflake-Labs/snowflakeR", ask = FALSE, upgrade = FALSE)
}

library(snowflakeR)
conn <- sfr_connect()

conn <- sfr_load_notebook_config(conn)

```

All table references should use fully qualified names via `sfr_fqn()`:

```
sfr_write_table(conn, sfr_fqn(conn, "MY_TABLE"), my_data)
```

See `vignette("workspace-notebooks")` for the full Workspace guide including output helpers, authentication details, and Python-R data exchange.

## Docker / CI / headless environments

For automated pipelines, containers, or GitHub Actions:

1. Install R and Python in your Dockerfile or CI step.
2. Use **key-pair authentication** (no browser available for SSO).
3. Set environment variables:

```

export SNOWFLAKE_ACCOUNT="xy12345.us-east-1"
export SNOWFLAKE_USER="SVC_USER"
export RETICULATE_PYTHON="/usr/bin/python3"

```

4. Or use a `connections.toml` mounted into the container.

For CI that only runs unit tests (no Snowflake connection needed), the Python environment just needs `snowflake-ml-python` installed so that `reticulate` can import the modules. The offline unit tests use mock objects.

See `.github/workflows/R-CMD-check.yml` in the repo for a working GitHub Actions example.

## 7. Optional dependencies

These R packages are optional but enhance `snowflakeR`:

| Package           | What it enables                                                                                                                                              |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RSnowflake        | DBI-compliant database access ( <code>dbGetQuery</code> , <code>dbWriteTable</code> , <code>dbplyr</code> , Connections Pane). See <code>RSnowflake</code> . |
| snowflakeauth     | <code>connections.toml</code> / <code>config.toml</code> credential management                                                                               |
| httr2             | Pure-R REST inference via <code>sfr_predict_rest()</code>                                                                                                    |
| knitr + rmarkdown | Building vignettes from source                                                                                                                               |

snowflakeR provides SQL querying via `sfr_query()` and `sfr_execute()` using the Python/Snowpark bridge. For full DBI compliance, `dbplyr` integration, and the RStudio Connections Pane, install `RSnowflake` and use `sfr_dbi_connection()` to bridge from an `sfr_connection`:

```
dbi_con <- sfr_dbi_connection(conn)
DBI::dbGetQuery(dbi_con, "SELECT 1")

library(dplyr)
tbl(dbi_con, "MY_TABLE") |> filter(score > 90) |> collect()
```

Install all optional dependencies at once:

```
pak::pak("Snowflake-Labs/snowflakeR", dependencies = TRUE)
```

## 8. Troubleshooting

| Issue                                                    | Solution                                                                                                     |
|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <code>reticulate</code> can't find Python                | Set <code>RETICULATE_PYTHON</code> in <code>.Renviron</code> or call <code>reticulate::use_condaenv()</code> |
| <code>snowflake.ml</code> import fails                   | Run <code>sfr_install_python_deps()</code> or <code>pip install snowflake-ml-python</code>                   |
| Python version mismatch                                  | Ensure Python $\geq 3.9$ ; check with <code>reticulate::py_config()</code>                                   |
| <code>sfr_connect()</code> hangs                         | If using <code>externalbrowser</code> , check your browser opened a Snowflake tab                            |
| <code>sfr_connect()</code> fails with "account required" | Set up <code>connections.toml</code> or pass <code>account</code> explicitly                                 |
| <code>reticulate</code> uses wrong Python                | Call <code>reticulate::use_condaenv()</code> <i>before</i> loading <code>snowflakeR</code>                   |
| R packages missing in Workspace                          | Add to YAML config or install interactively via <code>install.packages()</code>                              |
| <code>%R</code> not recognised in Workspace              | Run <code>%load_ext rpy2.ipynb</code> in a Python cell                                                       |
| Conda solve takes forever                                | Use <code>pip install</code> inside a conda env (see Option B above)                                         |

## Next steps

- `vignette("getting-started")` – Connecting, queries, DBI/dbplyr
- `vignette("model-registry")` – Logging and deploying R models
- `vignette("feature-store")` – Feature management and training data
- `vignette("workspace-notebooks")` – Full Workspace Notebook guide